

Minimum Time Consensus of Multi-agent System under Fuel Constraints

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Abstract:

This work addresses the problem of finding a consensus point in the state space for a multi-agent system that is comprised of N identical double integrator agents. It is assumed that each agent operates under constrained control input (i.e., $|u_i(t)| \leq 1 \forall i = 1, \dots, N$). Further, a fixed fuel budget is also assumed i.e., the total amount of cumulative input that can be expended is limited by $\int_0^{t_f} |u(t)| dt \leq \beta$. First, the attainable set $\mathcal{A}(t_f, \mathbf{x}_0, \beta)$ at time t_f , which is the set of all states that an agent can attain starting from initial conditions $\mathbf{x}_0$ under the fuel budget constraints at time t is computed for every agent. This attainable set is shown to be a convex set for all $t_f \geq 0$. Then the minimum time to consensus is the minimum time $\bar{t}_f$ at which attainable sets of all agents intersect, and the consensus point is the point of intersection. A closed-form expression for the minimum time consensus point is provided for the case of three agents. Then, using Helly's theorem, the intersection will be non-empty at a time when all the ${}^N C_3$ triplets of agents have non-empty intersection. The computation of minimum time consensus for all ${}^N C_3$ triplets is performed independently and can be distributed among all the N agents. Finally, the overall minimum time to consensus is given by the triplet that has the highest minimum time to consensus. Further, the intersection of all the attainable sets of this triplet gives the minimum time consensus point for all N agents.

1. INTRODUCTION

Decentralized control of a group of interconnected subsystems for various objectives such as consensus, synchronization, formation, etc has been studied extensively in the literature [Olfati-Saber et al. \(2007\)](#); [Ren and Cao \(2010\)](#). Practical applications of such objectives are mainly in the field of robotics [Fax and Murray \(2004\)](#), sensor networks [Cortés and Bullo \(2005\)](#), power networks, attitude alignment and consensus [Lynch \(1996\)](#), and so on. In a classical consensus problem, the control law is required such that each agent aims to match its state with that of the neighboring agent to reach a common agreement or point [Lin et al. \(2003\)](#); [Dong and Huang \(2013\)](#).

We consider a problem of computing minimum time consensus point for a group of double integrator agents with bounded control inputs and a limited fuel budget. The design of control laws to tackle consensus problems has already been well explored in the literature for both single-integrator and double-integrator agents, with the primary objective of achieving consensus [Ren and Beard \(2008\)](#). The speed of consensus can be adjusted through the connectivity. However, due attention

also needs to be paid to energy sources of these systems, driven by constraints on system resources such as operational time restrictions and fuel/energy usage. For example, in surveillance applications, UAVs are employed for monitoring and the success of mission relies on maximizing coverage within the constraints of limited fuel capacity. The critical aspect of fuel optimality in multi-agent systems context has recently been examined in [Sarkar et al. \(2021\)](#), focusing on finite time synchronization with minimizing time-fuel cost for double integrator agents interacting over a rooted directed tree.

Computation of control to achieve consensus in minimum time for double-integrator agents with complete communication has been explored in our earlier work [Mulla et al. \(2018, 2014\)](#), [Mulla et al. \(2018\)](#); [Patil and Mulla \(2019\)](#). However, the fuel budget associated with each agent's control input is also a crucial consideration for achieving consensus among N agents and remains unexplored to the best of our knowledge. Fuel budget becomes particularly crucial for extended UAV or UGV missions in remote areas where refueling poses challenges. This extended abstract deals with the computation of the minimum time consensus point for a group of N -agents while satisfying the constraints of a limited fuel budget associated with each agent.

* The work is partially supported by the SERB MATRICS grant MTR/2021/000882

2. PROBLEM FORMULATION

Consider N double integrator agents given by

$$\dot{\mathbf{x}}_i = A\mathbf{x}_i + Bu_i, \quad \mathbf{x}_i(0) = \mathbf{x}_{i0} \quad \text{for } i = 1, \dots, N \quad (1)$$

where, $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The system states are $\mathbf{x}_i = [x_i \ \dot{x}_i]^T \in \mathbb{R}^2$. The control input to i^{th} agent satisfies $|u_i(t)| \leq 1$ for all $t \geq 0$. The functional associated with i^{th} agent that describes the fuel consumption at time t_f is $F(u_i(t), t_f) = \int_0^{t_f} |u_i(t)| dt$ where, t_f is free. Let the maximum allowable fuel consumption be given as β ; then the *fuel budget constraint* is written as:

$$F(u_i(t), t_f) \leq \beta \quad (2)$$

Definition 1. A multi-agent system is said to achieve *consensus* if for all $i, j \in \{1, \dots, N\}$, where $i \neq j$, $\|\mathbf{x}_i(t) - \mathbf{x}_j(t)\| \rightarrow 0$ as $t \rightarrow \bar{t}_f > 0$ and $\mathbf{x}_i(t) = \mathbf{x}_j(t)$ for all $t \geq \bar{t}_f$. The time $\bar{t}_f$ is called the time to consensus, and the point $\mathbf{x}_i(\bar{t}_f) = \bar{\mathbf{x}}$ for $i = 1, \dots, N$ is the corresponding consensus point. The consensus is said to be achieved in finite time if $\bar{t}_f < \infty$.

Problem 1. Consider N agent system with given by (1). Find the minimum time to consensus $\bar{t}_f$ and the minimum time consensus point $\bar{\mathbf{x}} \in \mathbb{R}^2$ such that $\mathbf{x}_i(\bar{t}_f) = \mathbf{x}_j(\bar{t}_f) = \bar{\mathbf{x}}$ and $F(u_i(t), \bar{t}_f) \leq \beta$ for all $i, j \in \{1, 2, \dots, N\}$ with $i \neq j$.

Next we discuss the necessary preliminaries.

2.1 Preliminaries

Theorem 1. Athans and Falb (2013) For the system given by (1), there exists a unique bang-off-bang control $\bar{u}(t)$ that steers states from the initial conditions i.e., $\mathbf{x}_0 \in \bigcup_{t \geq 0} \mathcal{R}(t)$ to the origin with minimum fuel and in finite time t .

In light of Theorem 1, we consider the control profile for i^{th} agent with switching time instances as follows:

$$u_i(t) = \begin{cases} \gamma_1, & t \in [0, t_1] \\ 0, & t \in [t_1, t_2] \\ \gamma_2, & t \in [t_2, t_f] \end{cases} \quad 0 \leq t_1 \leq t_2 \leq t_f < \infty \quad (3)$$

where the sequences for $\{\gamma_1, 0, \gamma_2\}$ are: $s_1 : \gamma_1 = +1, \gamma_1 = -1, s_2 : \gamma_1 = 0, \gamma_1 = +1, s_3 : \gamma_1 = -1, \gamma_1 = +1, s_4 : \gamma_1 = 0, \gamma_2 = -1$. These control sequences incorporate a zero level, indicating that the signal remains inactive for a finite duration consuming no fuel. The fuel consumption upto time t_f is then $F(u(t_f), t_f) = t_f - t_2 + t_1$. Let the set of admissible control inputs be defined as

$$U = \{u_i(t) \text{ s.t. (3) and } F(u_i(t), t_f) \leq \beta\} \quad (4)$$

Definition 2. *Attainable Set* is the set of all the states that an agent can reach from $\mathbf{x}_{i0} \in \mathbb{R}^2$ using admissible control $u_i(t) \in U$ in time $t > 0$. The set $\mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta)$ from $\mathbf{x}_{i0}$ at time t_f is given by $\mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta) = \{e^{At_f} \mathbf{x}_{i0} + \int_0^{t_f} e^{A(t_f-\tau)} Bu_i(\tau) d\tau, \forall u_i(t) \in U\}$

In the above set, we have restricted $F(u(t_f), t_f)$ to be less than the β i.e., $t_f - t_2 + t_1 \leq \beta$. However, for characterizing full boundary of the set $\mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta)$ we

need to only consider the constraint $t_f - t_2 + t_1 = \beta$ which signifies the full utilization of the fuel budget. Further, also note that the set $\mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta)$ is a convex set.

2.2 Attainable Set Characterization

The states attained from $\mathbf{x}_{i0}$ at time t_f using the input profile (3) are

$$\mathbf{x}_i(t_f) = e^{At_f} \mathbf{x}_{i0} \pm \left(\int_0^{t_1} - \int_{t_2}^{t_f} \right) e^{A(t_f-\tau)} B d\tau. \quad (5)$$

Using the constraint $t_f - t_2 + t_1 = \beta$ gives the set $\mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta)$. For input sequence $\{+1, 0, -1\}$ we solve the equations obtained by expanding (5). For notational purpose we denote t_1, t_2 in this case as t_1^+, t_2^+ . Then relation between $\mathbf{x}_{i0} = [x_i, \dot{x}_i]$, t_1, t_2, t_f and β is as follows:

$$x_i - x_{i0} - \dot{x}_i t_f + \frac{(\dot{x}_i + \beta - \dot{x}_{i0})^2 + (\dot{x}_i - \beta + 2t_f - \dot{x}_{i0})^2}{8} \frac{t_f^2}{2} = 0 \quad (6)$$

Further, we get $t_1^+ = \frac{\dot{x}_i + \beta - \dot{x}_{i0}}{2}$ and $t_2^+ = \frac{\dot{x}_i - \beta + 2t_f - \dot{x}_{i0}}{2}$. Similarly, for control input $\{-1, 0, +1\}$ we get $t_1^- = \frac{-\dot{x}_i + \beta + \dot{x}_{i0}}{2}$ and $t_2^- = \frac{-\dot{x}_i - \beta + 2t_f + \dot{x}_{i0}}{2}$ and

$$x_i - x_{i0} - \dot{x}_i t_f - \left\{ \frac{(\dot{x}_i - \beta - \dot{x}_{i0})^2 + (\dot{x}_i + \beta + 2t_f - \dot{x}_{i0})^2}{8} \right\} + \frac{t_f^2}{2} = 0 \quad (7)$$

The set of points $\mathbf{x}_i$ that satisfies either equation (6) and (7) forms a part of the boundary $\partial \mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta)$ for the two control sequences. Let us denote these two parts by the following sets

$$\partial \mathcal{A}_i^+(t_f, \mathbf{x}_{i0}, \beta) := \left\{ \mathbf{x}_i \in \mathbb{R}^2 \mid (6) \text{ with } 0 \leq t_1^+ \leq t_2^+ \leq \infty \right\} \quad (8)$$

$$\partial \mathcal{A}_i^-(t_f, \mathbf{x}_{i0}, \beta) := \left\{ \mathbf{x}_i \in \mathbb{R}^2 \mid (7) \text{ with } 0 \leq t_1^- \leq t_2^- \leq \infty \right\} \quad (9)$$

For control input $\{0, +1\}$, considering $t_1 = 0$, we get

$$x_i = x_{i0} + \dot{x}_{i0} t_f - t_2 t_f + \frac{t_2^2}{2} + \frac{t_f^2}{2} \quad (10)$$

$$\dot{x}_i - \dot{x}_{i0} - \beta = 0. \quad (11)$$

Similarly, for control input $\{0, -1\}$, considering $t_1 = 0$, we get

$$x_i = x_{i0} + \dot{x}_{i0} t_f + t_2 t_f - \frac{t_2^2}{2} - \frac{t_f^2}{2} \quad (12)$$

$$\dot{x}_i - \dot{x}_{i0} + \beta = 0. \quad (13)$$

From equation (11) and equation (13), we see that this part of the boundary has fixed $\dot{x}_i$ co-ordinate. Thus,

$$\partial \mathcal{A}_i^{s_2} = \left\{ \mathbf{x}_i \in \mathbb{R}^2 \mid \dot{x}_i - \dot{x}_{i0} - \beta = 0, 0 \leq t_1 \leq t_2 < \infty \right\} \quad (14)$$

$$\partial \mathcal{A}_i^{s_4} = \left\{ \mathbf{x}_i \in \mathbb{R}^2 \mid \dot{x}_i - \dot{x}_{i0} + \beta = 0, 0 \leq t_1 \leq t_2 < \infty \right\} \quad (15)$$

The boundary of the set $\mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta)$ is the union of (8), (9), (14), and (15). An example is shown in Figure 1.

3. MINIMUM TIME CONSENSUS COMPUTATION

Now note that for fixed β , in the $(x, \dot{x})$ -plane the set $\mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta)$ grows unbounded only in x direction as

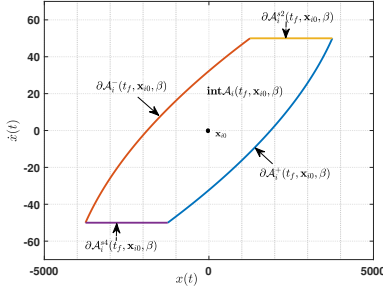


Fig. 1. $\partial\mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta)$ with $\mathbf{x}_{i0} = [0, 0]^T$, $\beta = 50$ and $t_f = 100$

$t_f \rightarrow \infty$. In the $\dot{x}$ direction the set is bounded in both directions for all $t_f \in [0, \infty)$. Thus, the consensus is possible only for a certain set of initial conditions. We assume that all the N agents have initial conditions so that consensus is possible. Next, we note that $\mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta)$ is a convex set. Thus, to compute the minimum time consensus point we make use of the Helly's theorem stated next.

Theorem 2. Helly's Theorem Gruber (2007): Let F be a finite family of at least $n + 1$ convex sets on $\mathbb{R}^n$. If the intersection set of every $n + 1$ members of F is non-empty, then all members of F have a non-empty intersection.

For our case $n = 2$. Therefore, the problem of computing consensus point can be broken into computing consensus of only a triplet of agents.

3.1 Minimum time consensus for $N = 3$

Consider a triplet of agents $\{a_i, a_j, a_k\}$. We need to compute the minimum time t_f such that the intersection of the sets $\mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta)$, $\mathcal{A}_j(t_f, \mathbf{x}_{j0}, \beta)$ and $\mathcal{A}_k(t_f, \mathbf{x}_{k0}, \beta)$ is non empty. Hence, for every pair i.e., $(i, j), (j, k)$ and (i, k) , their attainable sets must intersect first. Therefore, $\bar{t}_{ijk} \geq \max\{\bar{t}_{ij}, \bar{t}_{jk}, \bar{t}_{ik}\}$. Here, $\bar{t}_{ij}$ is the minimum time t_f such that $\mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta) \cap \mathcal{A}_j(t_f, \mathbf{x}_{j0}, \beta) \neq \emptyset$. Without loss of generality, assume $\bar{t}_{ij} = \max\{\bar{t}_{ij}, \bar{t}_{jk}, \bar{t}_{ik}\}$. We now have two cases: 1) $\bar{\mathbf{x}}_{ij} \in \mathcal{A}_k(\bar{t}_{ij}, \mathbf{x}_{k0}, \beta)$ and 2) $\bar{\mathbf{x}}_{ij} \notin \mathcal{A}_k(\bar{t}_{ij}, \mathbf{x}_{k0}, \beta)$

For case 1, we just need to compute the solution to the expressions defining the boundary of two attainable sets (see Section 2.2). We discuss case 2 which requires significant computation. In case 2, for each pair of agents, the corresponding attainable sets intersect at time $\bar{t}_{ij} = \max\{\bar{t}_{ij}, \bar{t}_{jk}, \bar{t}_{ik}\}$. For the existence of consensus point among three agents (denoted by $\bar{\mathbf{x}}_{ijk}$), the attainable sets of every pair must intersect with each other. For finding first consensus point, the point must exist at some time t_f on the boundaries of attainable sets. Therefore, the following condition must hold true for $\bar{t}_f \leq t_f$ as:

$$\partial\mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta) \cap \partial\mathcal{A}_j(t_f, \mathbf{x}_{j0}, \beta) \cap \partial\mathcal{A}_k(t_f, \mathbf{x}_{k0}, \beta) \neq \emptyset. \quad (16)$$

Checking this condition requires computing solution of set of polynomial equations with inequality constraints simultaneously. We assume that $\bar{\mathbf{x}} \in \partial\mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta) \cap \partial\mathcal{A}_j(t_f, \mathbf{x}_{j0}, \beta) \cap \partial\mathcal{A}_k(t_f, \mathbf{x}_{k0}, \beta)$. Since $\bar{\mathbf{x}}_{ijk}$ lie on the boundary of attainable sets, control inputs of the form (3) are required to drive the states of all agents to $\bar{\mathbf{x}}_{ijk}$

at time $\bar{t}_{ijk}$. Therefore, depending upon the structure of the input, there are various ways in which $\bar{\mathbf{x}}_{ijk}$ is the element of the attainable sets. We consider boundaries determined by quadratic curves in $\mathbb{R}^2$, namely $\partial\mathcal{A}_i^+(t_f, \mathbf{x}_{i0}, \beta)$ and $\partial\mathcal{A}_i^-(t_f, \mathbf{x}_{i0}, \beta)$. At time $t_f = \bar{t}_{ijk}$, these sets meet each other at a unique point. To enumerate all possible combinations in which the boundaries described by (6) and (7) intersect are outlined below. For brevity we omit the arguments while writing the sets. $\partial\mathcal{A}_i^+ \cap \partial\mathcal{A}_j^+ \cap \partial\mathcal{A}_k^+$, $\partial\mathcal{A}_i^+ \cap \partial\mathcal{A}_j^+ \cap \partial\mathcal{A}_k^-$, $\partial\mathcal{A}_i^+ \cap \partial\mathcal{A}_j^- \cap \partial\mathcal{A}_k^+$, $\partial\mathcal{A}_i^+ \cap \partial\mathcal{A}_j^- \cap \partial\mathcal{A}_k^-$, $\partial\mathcal{A}_i^- \cap \partial\mathcal{A}_j^+ \cap \partial\mathcal{A}_k^+$, $\partial\mathcal{A}_i^- \cap \partial\mathcal{A}_j^+ \cap \partial\mathcal{A}_k^-$, $\partial\mathcal{A}_i^- \cap \partial\mathcal{A}_j^- \cap \partial\mathcal{A}_k^+$, $\partial\mathcal{A}_i^- \cap \partial\mathcal{A}_j^- \cap \partial\mathcal{A}_k^-$. The possibility of all agents having the same sign is excluded i.e., $\bar{\mathbf{x}}_{ijk}$ will not be in the intersection of sets with the same signs. There exists few more cases where the intersection point lies on one of the agent's boundary described by (11) and (13). In each of the six cases, we solve three equations corresponding to agents a_i, a_j, a_k . This results in six closed-formulas expressing t_f in terms of x_{i0}, x_{j0} , and x_{k0} , denoted as $t_1, \dots, t_6$ respectively. We obtained quadratic equation in t_f i.e., $at_f^2 + bt_f + c = 0$ where,

$$\begin{aligned} a &= \dot{x}_{j0}^2 - 2\dot{x}_{j0}\dot{x}_{k0} + \dot{x}_{k0}^2, \\ b &= -2\dot{x}_{i0}\dot{x}_{j0}^2 + 4\dot{x}_{i0}\dot{x}_{j0}\dot{x}_{k0} - 2\dot{x}_{i0}\dot{x}_{k0}^2 + 4x_{j0}\dot{x}_{j0} - 4x_{j0}x_{k0} \\ &\quad + \dot{x}_{j0}^3 - \dot{x}_{j0}^2\dot{x}_{k0} - 2\dot{x}_{j0}^2\beta - 4\dot{x}_{j0}x_{k0} - \dot{x}_{j0}\dot{x}_{k0}^2 + 4\dot{x}_{j0}x_{k0}\beta \\ &\quad + 4x_{k0}\dot{x}_{k0} + \dot{x}_{k0}^3 - 2\dot{x}_{k0}^2\beta, \\ c &= -2x_{i0}\dot{x}_{j0}^2 + 4x_{i0}\dot{x}_{j0}\dot{x}_{k0} - 2x_{i0}\dot{x}_{k0}^2 + 0.5\dot{x}_{i0}^2\dot{x}_{j0}^2 - \dot{x}_{i0}^2\dot{x}_{j0}\dot{x}_{k0} \\ &\quad + 0.5\dot{x}_{i0}^2\dot{x}_{k0}^2 - 2\dot{x}_{i0}x_{j0}\dot{x}_{j0} + 2\dot{x}_{i0}x_{j0}\dot{x}_{k0} - 0.5\dot{x}_{i0}\dot{x}_{j0}^3 \\ &\quad + 0.5\dot{x}_{i0}\dot{x}_{j0}^2\dot{x}_{k0} + 2\dot{x}_{i0}\dot{x}_{j0}x_{k0} + 0.5\dot{x}_{i0}\dot{x}_{j0}\dot{x}_{k0}^2 - 2\dot{x}_{i0}x_{k0}\dot{x}_{k0} \\ &\quad - 0.5\dot{x}_{i0}\dot{x}_{k0}^3 + 4x_{j0}^2 + 2x_{j0}\dot{x}_{j0}^2 - 2x_{j0}\dot{x}_{j0}\dot{x}_{k0} - 8x_{j0}x_{k0} \\ &\quad + 0.25\dot{x}_{j0}^4 - 0.5\dot{x}_{j0}^3\dot{x}_{k0} + 0.5\dot{x}_{j0}^2\dot{x}_{k0}^2 + \dot{x}_{j0}^2\beta^2 + 2\dot{x}_{j0}x_{k0}\dot{x}_{k0} \\ &\quad - 0.5\dot{x}_{j0}\dot{x}_{k0}^3 - 2\dot{x}_{j0}\dot{x}_{k0}\beta^2 + 4x_{k0}^2 + 2x_{k0}\dot{x}_{k0}^2 + 0.25\dot{x}_{k0}^4 \\ &\quad + \dot{x}_{k0}^2\beta^2 \end{aligned} \quad (17)$$

Remark 1. The analytical expressions for $\bar{t}_f$, $\bar{x}$ and $\bar{\dot{x}}$ are computed using the Gröbner Basis Cox et al. (2013). The computation of the Gröbner basis in the present work was performed using the software package described in Stein (2007).

The solution obtained for each of these cases is denoted by $t_{fl}, l = 1, \dots, 6$. The corresponding values of intersection point denoted by $\bar{\mathbf{x}}(t_{fl})$ is determined by solving the three equations associated with that specific case. The closed-form expression for $l = 1$, i.e., $\bar{\mathbf{x}}(t_{f1})$ is given as:

$$\begin{bmatrix} \bar{x}(t_{f1}) \\ \bar{\dot{x}}(t_{f1}) \end{bmatrix} = \begin{bmatrix} \frac{2\dot{x}(t_{f1})(2\bar{t}_{f1} - \dot{x}_{i0} + \dot{x}_{k0}) + 2\bar{t}_{f1}(\dot{x}_{i0} + \dot{x}_{k0}) + 4(x_{i0} + x_{k0}) + \dot{x}_{i0}^2 - \dot{x}_{k0}^2}{-2x_{j0} + 0.5\dot{x}_{j0}^2 + 2x_{k0} - 0.5\dot{x}_{k0}^2} \\ -\bar{t}_{f1} + \frac{\dot{x}_{j0} - \dot{x}_{k0}}{\dot{x}_{j0} - \dot{x}_{k0}} \end{bmatrix}$$

Similar expressions for $\bar{\mathbf{x}}(t_l), l = 2, \dots, 6$ can be obtained. Now, from the value of t_l and $\bar{\mathbf{x}}(t_l)$, the expressions for the switching time instances i.e., t_{s1} and t_{s2} for each agent $\alpha \in \{i, j, k\}$ are obtained. In computing the set l , if $\partial\mathcal{A}_\alpha^+$ or $\partial\mathcal{A}_\alpha^-$ is used, then the switching time instances for a_α for both cases are given as: $t_{1s\alpha l}^+ = \frac{\dot{x}_i(t_{fl}) + \beta - \dot{x}_{i0}}{2}$, $t_{2s\alpha l}^+ = \frac{\dot{x}_i(t_{fl}) - \beta + 2t_{fl} - \dot{x}_{i0}}{2}$, $t_{1s\alpha l}^- = \frac{-\dot{x}_i(t_{fl}) + \beta + \dot{x}_{i0}}{2}$ and $t_{2s\alpha l}^- = \frac{-\dot{x}_i(t_{fl}) - \beta + 2t_{fl} + \dot{x}_{i0}}{2}$. The set l for which the inequality $0 \leq t_{1s\alpha l} \leq t_{2s\alpha l} \leq t_f < \infty$ holds for all $\alpha = i, j, k$, we have $\bar{t}_{ijk} = t_l$. The corresponding value of $\bar{\mathbf{x}}(t_l)$ is the consensus point $\bar{\mathbf{x}}_{ijk}$.

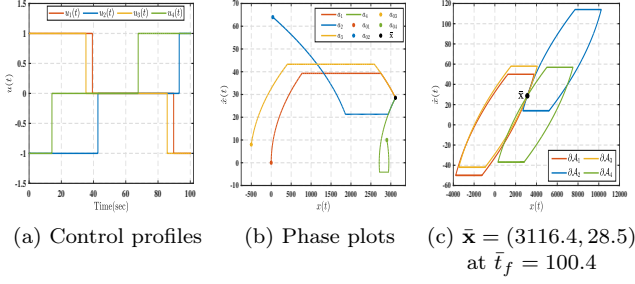


Fig. 2. For $N = 4$ agents in $\mathbb{R}^2$

3.2 Minimum time consensus for N agents

From Helly's theorem, it follows that $\bar{t}_f = \max_{1 \leq i,j,k \leq N} \bar{t}_{ijk}$ if and only if attainable set of all ${}^N C_3$ triplets of agents have non-zero intersections at $\bar{t}_f$.

Algorithm 1 Minimum time consensus under fixed fuel budget

Inputs: $\mathbf{x}_{i0}$, β for $i = 1, \dots, N$
Outputs: $\bar{t}_f$, $\bar{\mathbf{x}}$
for $i = 1, 2, \dots, N$ **do**
 Generate all possible triplets (say m)
end for
for $j = 1, 2, \dots, m$ **do**
 for $k = 1, 2, \dots, 6$ **do**
 Compute $\bar{t}_{fk}$ using equation (17).
 Compute $\bar{\mathbf{x}}_k$ by solving equations (6) and (7) simultaneously.
 end for
 $\bar{t}_{fk} = \{\bar{t}_{f1}, \dots, \bar{t}_{f6}\}$
 $\bar{t}_{fj} = \min\{\bar{t}_{f1}, \dots, \bar{t}_{f6}\}$
end for
 $t = \{\bar{t}_{f1}, \dots, \bar{t}_{fm}\}$
 $\bar{t}_f = \max\{t\}$
 $\bar{\mathbf{x}} = \mathbf{x}$ at $\bar{t}_f$

4. EXAMPLE

Consider a set of initial conditions for four agents, i.e., $\mathbf{x}_{10} = [0, 0]^T$, $\mathbf{x}_{20} = [40, 64]^T$, $\mathbf{x}_{30} = [-500, 8]^T$, $\mathbf{x}_{40} = [2900, 10]^T$. Also, let the associated fuel budget with each agent be $\beta = 50$. There are 4 triplets of agents formed which are $\{a_1, a_2, a_3\}$, $\{a_1, a_2, a_4\}$, $\{a_1, a_3, a_4\}$ and $\{a_2, a_3, a_4\}$. The minimum time to consensus is maximum for $\{a_1, a_2, a_3\}$ triplets. For $\{a_1, a_2, a_3\}$ triplet, the consensus point belongs to the boundaries of attainable set i.e., $\bar{\mathbf{x}} \in \partial \mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta)$. For remaining agent i.e a_4 , $\bar{\mathbf{x}} \in \text{int} \mathcal{A}_i(t_f, \mathbf{x}_{i0}, \beta)$, now to reach the consensus point in $\bar{t}_f$, the fuel budget is recomputed by substituting the minimum time to consensus and the corresponding consensus states in (6) and (7) which turned out to be less than initial specified fuel budget. The updated fuel budget is computed as $\beta' = 46.9$. Also, time instances computed with the updated β' are $t_1 = 14.2$ and $t_2 = 67.7$ for control profile $\{-1, 0, +1\}$. The simulation results are shown in Figure

2 to demonstrate the minimum time consensus with a specified fuel budget.

5. CONCLUSIONS

The minimum time consensus scheme considered in this work requires all to all agent communication. Future work includes relaxing the connectivity assumption and developing a decentralized control scheme for achieving the minimum time consensus with fixed fuel budget.

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